University of East Anglia Norwich

School of Computing Sciences

Advanced Web Development

CMP-6057A & CMP-7057A

Jeannette Chin

Session Schedule

- Module information
  - Module structure, team, teaching approach and topics, resource required and provided, and reading list
- Assessment information
  - Assessment elements, timeline, and submission requirements

Module Information

Teaching Team and Office Hours



Module lecturer

Dr Jeannette Chin

- Office: Sci2.18



Module Organiser & Lecturer

Dr Cheng Wang

- Office: Sci2.30

Lab tutors:

Amber Yucebas
A.Yucebas@uea.ac.uk

Darcey Gardiner
Darcey.Gardiner@uea.ac.uk

What this module is about ?

The module covers full stack Next.js with REACT > v19 web development and PostgresSQL DB integration, Web security, WebAI transformers.js, tensorflow.js on pre-trained model, customed model and transfer learning.

By completing the module, you will gain an understanding of:

- Full stack web development using Next.js & REACT
- Typescript, API programming
- Web Security implementation and PostgresSQL DB integration
- Deploy ML models on a browser

- Gain more experience of group work / team project / demo / presentation

What this module is about ?

Some weeks you may find there are a lot of lab exercises …

- this is intended.

- can’t finish them in the lab – finish them as homework

- each lab exercise is linked to the previous one, so there will be a lot of homework for you in the first 5 weeks.

- to make the most of this module, try finish the lab exercises each week if you can

What this module is not about ?

This module is NOT about learning everything about Next or REACT

- we will cover as much as possible within the given time frame

This module is NOT about learning everything about WebAI

- again, we will cover all much as possible, all the useful and basics.

This module is NOT about learning hosting server engineering / security configuration / load balancing / backups /

- out of scope

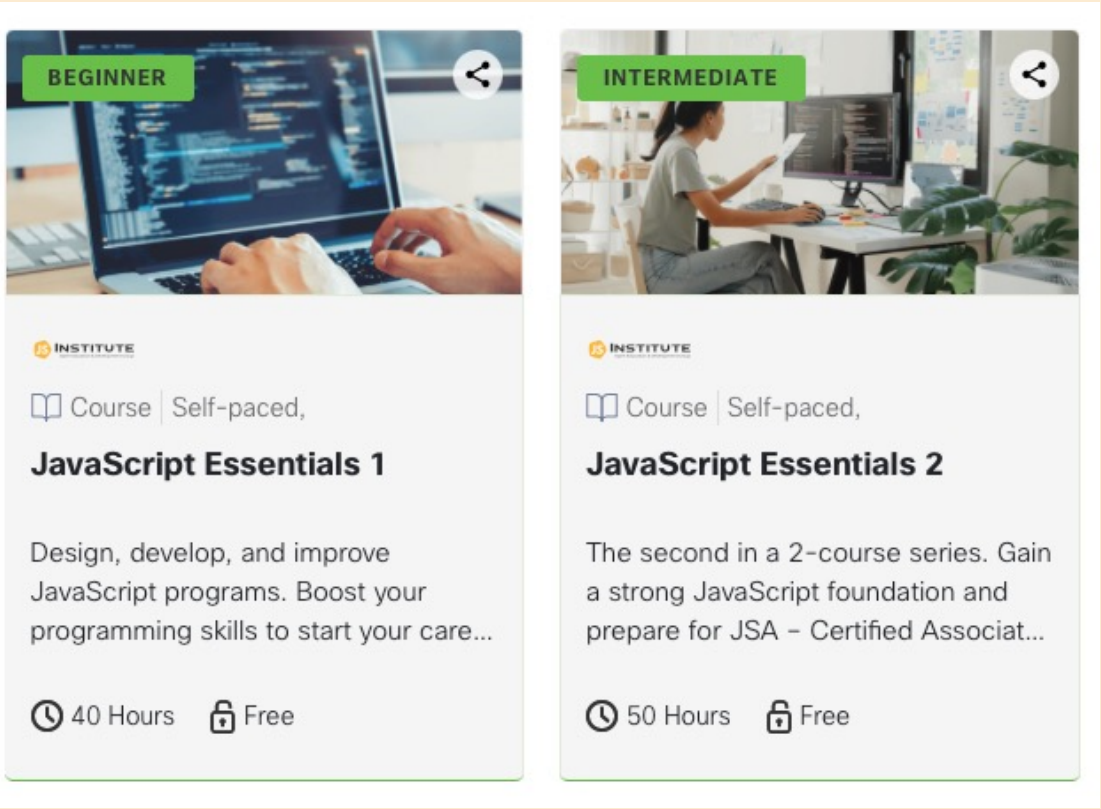
By the end of the module, you will have gained the WebDev skills and confidently build industry-standard apps and keep learning and exploring

JavaScript

- I am assuming you are an experienced JS programmer
  - JS function
  - Async programming

Anyone interested in JS programming self-paced course?

Email me!



Teaching Schedule

CMP-6057A and CMP-7057A Advanced Web Development 2026/27					
	Lectures	**Lecturer**	**Laboratory**		**Assessment**
Week 1	Introduction to the module, Introduction to Typescript	Jeannette Chin	Typescript exercises		CW1 release
Week 2	Reusable components with React	Jeannette Chin	React lab exercises		
Week 3	Next.js, global state management & rendering strategy, API Programming	Jeannette Chin	Next.js lab exercises		
Week 4	Web Security with Auth.js & Prisma ORM	Jeannette Chin	Auth.js & Prisma lab exercises		
Week 5	Custom Credentials, Sessions Management with full stack	Jeannette Chin	Full stack Custom Credential lab exercises		
Week 6	Independent Learning Week		CW1 catch up		
Week 7	Web AI foundations with TensorFlow.js	Cheng Wang	TensorFlow.js lab exercises		CW1 submission + CW2 release
Week 8	Pretrained AI models with Transformers.js	Cheng Wang	Transformers.js lab exercise		CW1 viva sessions
Week 9	WebAI applications with Transformers.js	Cheng Wang	Transformers.js lab exercise		
Week 10	AI-assisted workflows and bounded web agents	Cheng Wang	Local-model workflow exercise		
Week 11	Tool use, MCP and AI integration for web applications	Cheng Wang	MCP lab exercise		
Week 12	Catch up	Cheng Wang	CW2 catch up		
CW2 due 18/01/2027 Viva ??					



Lectures and labs

- Weekly 2-hour lecture and 2-hour lab in a block schedule on Tuesdays
  - 14:00 – 18:00
  - In Room: Sci 0.59

Module Assessment

- Two assignments (both individual):
  - 001 Web Application Project 60%
    - Video demonstration (.mp4) & source code
    - CMP-7057A only – a group technical report (max 5 pages)
    - 6 November 2026, 3pm (week 7)
    - In person VIVA sessions – Weds &Thurs week 8
  - 002 Implementing WebAI 40%
    - Video demonstration (.mp4) & source code
    - 18 January 2027, 3pm
    - In person VIVA sessions – TobeConfirmed



Assignment 001 is out!

- Check it out on Blackboard.

Passing the module

- UG / Y3 – All you need to do is to obtain an overall mark of 40% or above
- Y4 / PGT - All you need to do is to obtain an overall mark of 50% or above.



Summary

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